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DATABASE SYSTEM MANAGEMENT

MIDTERM REPORT

Topic: Tour Guide Service Management

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Repository: https://github.com/ttkien0710/Group12_CSDL_BA65

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A guide fee and a payment answer different questions. The fee records what the customer owes for a completed trip. A payment records how much the operator actually received on a particular date. Keeping the two separate lets the operator track instalments and unpaid trips while still reporting which guides led each group. The database covers guide services only. Transport, accommodation and individual passenger records are outside its scope.

A. Database design

A.1 How the records fit together

Each trip has one customer representing the group and one guide responsible for the itinerary. The destination and group size belong to the trip. The guide's operating region, by contrast, is recorded once in guides, so trips can be grouped by region through their assigned guide. The two tables below trips in Figure 1 reflect different relationships: a completed trip has a single report to confirm the fee, but the customer may settle it in several payments.

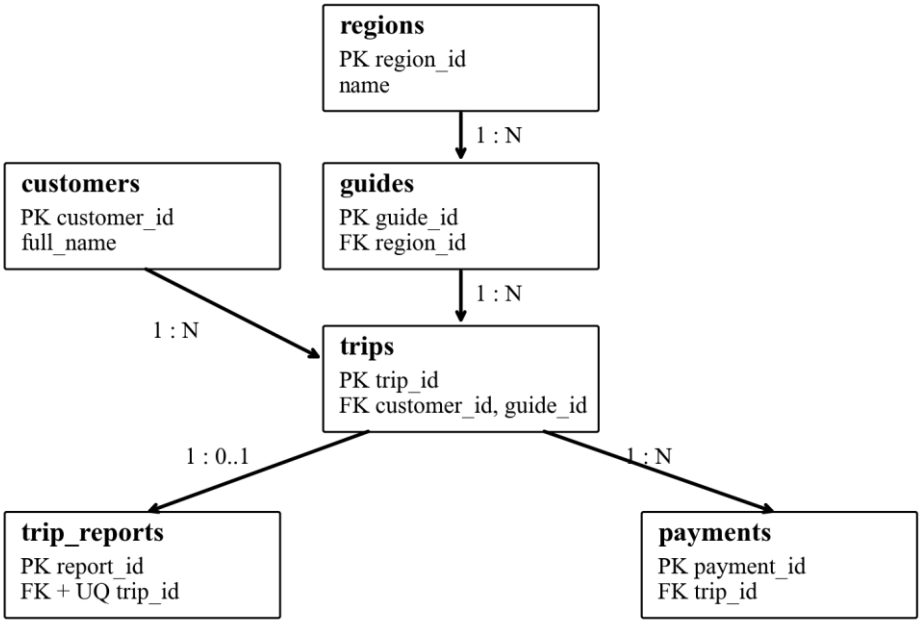


Figure 1. Relationships among the six tables. PK = primary key, FK = foreign key, UQ = unique constraint. A region, guide or customer can exist before it has any trips associated with it.

A.2 Table definitions and integrity rules

schema.sql creates six InnoDB tables in tour_guide_mini. Each table begins with its primary key. The foreign keys link records by ID, which matters here because customers 1 and 10 are both named Nguyễn Văn An.

Table	Primary key	Other fields
regions	region_id	name
guides	guide_id	full_name, email, language, region_id
customers	customer_id	full_name, phone, email

Table	Primary key	Other fields
trips	trip_id	customer_id, guide_id, departure_date, status, destination, participant_count
trip_reports	report_id	trip_id, finished_at, itinerary_summary, feedback, fee
payments	payment_id	trip_id, amount, payment_date, method

A foreign key on trip_reports.trip_id requires every report to refer to an existing trip. Making the same column unique prevents two reports from confirming a fee for that trip. By design, payments.trip_id is allowed to repeat: two receipts can share a trip ID and still be separate transactions with different payment IDs.

Names, business dates, required trip references and monetary amounts are mandatory. Dates use DATE, and both fees and receipts use DECIMAL(14,2). A trip must have at least one participant and one of three statuses: Pending, Completed or Cancelled. A confirmed fee may be zero. Each receipt, however, must record a positive amount paid by Cash, BankTransfer or Card.

Uniqueness sets an upper limit of one report per trip. It cannot establish that every Completed trip has a report, or that a Pending or Cancelled trip has none. The status check in Part B tests both conditions against the loaded data.

Two further rules involve several records: a receipt needs a confirmed fee, and the receipts for a trip may not exceed it. schema.sql ends with an optional BEFORE INSERT trigger on payments that rejects a receipt for a trip without a report, or one that would push the total above the fee.

B. Sample data and integrity checks

B.1 Sample records

seed.sql inserts regions, guides and customers before the trips that reference them. Reports and receipts follow the trips. The fictional records meet the minimum counts below and include cases that distinguish a valid zero from a missing record.

Table	Minimum required	Records loaded
regions	5	5
guides	6	6
customers	10	10
trips	12	17
trip_reports	10	13
payments	10	13

Examples from the sample records

Requirement	Example from the sample data

Requirement	Example from the sample data
Dates inside and outside the reporting period	Trips 1 and 2 departed in January 2025, trip 3 in February and trip 4 in March. Trip 7 departed in April 2025 and trip 10 in December 2024.
All three trip statuses	Of 17 trips, 13 are Completed, two are Pending (11 and 13) and two are Cancelled (12 and 14).
More than three completed trips in 2025	Customer 1 completed trips 1–4. Customer 10 has the same name but is a separate customer with one completed trip (16).
Trip history without payment	Customer 3 has a confirmed fee of VND 700,000 for trip 6 and has made no payments.
Multiple receipts and an outstanding balance	For trip 1, receipts of VND 400,000 and VND 300,000 leave VND 300,000 of the VND 1,000,000 fee outstanding. Trip 16 was also paid in two receipts (VND 200,000 and VND 300,000) and is settled in full.
Zero activity in the specified period	Regions 4 and 5 have no trips in Q1 2025. Guide 6 has no trips in 2025 but led trip 10 in 2024.
A valid zero fee	Trip 17 is Completed with a confirmed fee of VND 0 and no payment, so its balance is zero rather than missing.

B.2 Validation results

For these records, finished_at marks the date the report confirms the fee. Each check in seed.sql returns one summary row with its result (OK or FAIL) and the number of rows that break the rule. All five returned OK with zero violations.

Validation query	Violation sought	Violations
Foreign keys	An orphan reference in any of the five parent–child relationships.	0
Status and report	A Completed trip with other than one report, or a Pending or Cancelled trip with a report.	0
Completion date	A report whose finished_at precedes the trip's departure_date.	0
Fee and receipts	A payment for a trip without a confirmed fee, or receipts whose sum exceeds that fee.	0
Receipt date	A payment recorded before the report's finished_at date.	0

Primary keys, foreign keys, unique constraints and row-level checks apply when data are written, and the payments trigger blocks a receipt that has no confirmed fee or exceeds it. The checks above establish whether the sample satisfies rules that depend on several records. They can be rerun after a change, for example to audit total payments against each trip fee after data are loaded by another route.

B.3 Running the assignment

schema.sql drops and recreates tour_guide_mini, so the scripts can be rerun from the start. Run them in this order: schema.sql, seed.sql, queries.sql and views.sql. The third script reads the base tables directly, so all ten queries can run before the views are created. The trigger at the end of schema.sql uses DELIMITER, so run the scripts with the mysql client or MySQL Workbench.

The statements ran on a fresh MySQL 8.0.46 database with STRICT_TRANS_TABLES and ONLY_FULL_GROUP_BY enabled. Parts C and D use the results of that run. All monetary amounts are in Vietnamese đồng (VND).

C. Ten queries and their results

The numbering follows queries.sql. Trip activity is dated by departure_date. Receipts are dated by payment_date.

C1. Completed trips in January 2025

Joining each completed January 2025 trip to its customer and guide identifies who booked the trip and who led it. The date shown is the departure date.

Trip	Customer ID	Customer	Guide ID	Guide	Departure
1	1	Nguyễn Văn An	1	Nguyễn Minh Anh	2025-01-05
2	1	Nguyễn Văn An	2	Trần Quốc Bảo	2025-01-20

C2. Trips by region in the first quarter of 2025

Regions 4 and 5 had no trips in Q1 2025. To keep them in the count, the LEFT JOIN to trips applies the date range in its ON clause, from 1 January up to but excluding 1 April. Applying that range in WHERE would discard the rows with no matching trip. COUNT(t.trip_id) then gives those regions zero. COUNT(*) would count the outer-join row itself. All trip statuses count toward the quarterly total.

Region ID	Region	Trips
1	Miền Bắc	5
2	Bắc Trung Bộ	3
3	Nam Trung Bộ	1
4	Tây Nguyên	0
5	Nam Bộ	0

C3. Customers with an unreported trip that was not cancelled

Customers 2 and 8 qualify through pending trips 13 and 11 respectively. EXISTS tests for at least one non-cancelled trip without a report, so a customer with several such trips would still appear only once. Customer 9 has only a cancelled trip and is excluded.

Customer ID	Customer
2	Trần Thị Bình
8	Bùi Ngọc Mai

C4. Average confirmed fee per completed trip by region

The unit of the average is a completed trip. Each trip contributes one confirmed fee, regardless of how many payments the customer made. Joining only completed trips with reports also limits the output to regions with a confirmed fee. Trip 17 has a confirmed fee of zero and is included, which brings the average for region 2 down to VND 500,000.

Region ID	Region	Average fee (VND)
1	Miền Bắc	940,000
2	Bắc Trung Bộ	500,000
3	Nam Trung Bộ	1,300,000
4	Tây Nguyên	900,000
5	Nam Bộ	1,000,000

Unlike the quarterly count in C2, this average includes all dates. Region 5 therefore qualifies through its trip in December 2024.

C5. Three guides with the most completed trips in 2025

Guides 1 and 3 lead with three completed trips each in 2025. Guides 2, 4 and 5 tie at two. Sorting first by count and then by ascending guide ID gives the third place to guide 2, and guides 4 and 5 fall outside the cutoff despite having the same count.

Guide ID	Guide	Completed trips
1	Nguyễn Minh Anh	3
3	Lê Thu Hà	3
2	Trần Quốc Bảo	2

C6. Payments received from each customer in 2025

For cash receipts, the relevant year is the year the money arrived. Filtering payment_date counts payments made in 2025 even if the trip departed in another year. Starting with customers and applying the date filter in the payment join keeps customers who paid nothing during the year.

Customer ID	Customer	Receipts in 2025 (VND)
1	Nguyễn Văn An	4,200,000
2	Trần Thị Bình	900,000
3	Lê Minh Châu	0

Customer ID	Customer	Receipts in 2025 (VND)
4	Phạm Quốc Dũng	1,100,000
5	Hoàng Thu Giang	2,300,000
6	Vũ Đức Hải	600,000
7	Đặng Mỹ Linh	0
8	Bùi Ngọc Mai	0
9	Đỗ Thanh Nam	0
10	Nguyễn Văn An	500,000

Customer 7 illustrates why an annual total of zero does not establish an unpaid balance. The customer settled the fee in December 2024.

C7. Customers who have never made a payment

NOT EXISTS checks the customer's entire trip history for a receipt. This separates customers who have never paid from those who simply paid in a different year.

Customer ID	Customer
3	Lê Minh Châu
8	Bùi Ngọc Mai
9	Đỗ Thanh Nam

Customer 3 has an unpaid completed trip. Customer 8 has only a pending trip and a completed trip with a zero fee, and customer 9 has only a cancelled trip. All three belong in this list. Customer 10 is not listed because the receipts for trip 16 count as payments.

C8. Customers ranked by total confirmed fees

The ranking uses confirmed fees, so customer 3's unpaid VND 700,000 still counts. Customers with no confirmed fees remain in the ranking with zero. Ties are ordered by ascending customer ID.

Customer ID	Customer	Confirmed fees (VND)
1	Nguyễn Văn An	4,500,000
5	Hoàng Thu Giang	2,300,000
4	Phạm Quốc Dũng	1,100,000
7	Đặng Mỹ Linh	1,000,000
2	Trần Thị Bình	900,000
3	Lê Minh Châu	700,000
6	Vũ Đức Hải	600,000
10	Nguyễn Văn An	500,000
8	Bùi Ngọc Mai	0
9	Đỗ Thanh Nam	0

C9. Cancellation rate by guide in 2025

For each guide, the denominator includes every trip departing in 2025, across all statuses. The numerator counts cancellations. The percentage is rounded to two decimal places.

Guide ID	Guide	All trips	Cancelled	Rate (%)
1	Nguyễn Minh Anh	4	0	0.00
2	Trần Quốc Bảo	3	1	33.33
3	Lê Thu Hà	4	1	25.00
4	Phạm Đức Huy	2	0	0.00
5	Hoàng Ngọc Lan	3	0	0.00
6	Vũ Hải Nam	0	0	NULL

Guide 1 led four trips and none was cancelled, giving 0.00%. Guide 3 led four trips and one was cancelled, giving 25.00%. Guide 6 had no trips in 2025. NULLIF converts that zero denominator to NULL, so the report leaves the rate undefined.

C10. Customers with more than three completed trips in 2025

After filtering for Completed departures in 2025, GROUP BY counts trips for each customer ID and name. HAVING retains counts strictly greater than three and the result is ordered by decreasing count. Grouping by ID keeps the two customers named Nguyễn Văn An distinct.

Customer ID	Customer	Completed trips
1	Nguyễn Văn An	4

D. Reporting views

D1. Trip details (v_detail)

v_detail keeps one row per trip. A LEFT JOIN brings in the fee when a completion report exists. The six selected rows below illustrate the difference between completed, pending and cancelled trips, and between a zero fee and a missing one. The full view contains 17 rows and includes the customer, guide and region IDs as well as their names.

Trip	Departure	Status	Customer	Guide	Region	Fee (VND)
1	2025-01-05	Completed	Nguyễn Văn An	Nguyễn Minh Anh	Miền Bắc	1,000,000
2	2025-01-20	Completed	Nguyễn Văn An	Trần Quốc Bảo	Miền Bắc	1,200,000
10	2024-12-15	Completed	Đặng Mỹ Linh	Vũ Hải Nam	Nam Bộ	1,000,000
11	2025-03-28	Pending	Bùi Ngọc Mai	Nguyễn Minh Anh	Miền Bắc	NULL
12	2025-02-22	Cancelled	Đỗ Thanh Nam	Lê Thu Hà	Bắc Trung Bộ	NULL
17	2025-08-05	Completed	Bùi Ngọc Mai	Lê Thu Hà	Bắc Trung Bộ	0

For trip 11, NULL means the fee has yet to be confirmed. Trip 17 shows the other case: its report confirms a fee of zero, so the view shows 0 rather than NULL.

D2. Regional activity (v_group_summary)

v_group_summary counts trips of every status and gives separate counts for Completed and Cancelled trips. It starts from regions so that a region with no trips still appears with zero trips and zero total fees.

Region ID	Region	All trips	Completed	Cancelled	Fees (VND)
1	Miền Bắc	7	5	1	4,700,000
2	Bắc Trung Bộ	4	3	1	1,500,000
3	Nam Trung Bộ	2	2	0	2,600,000
4	Tây Nguyên	3	2	0	1,800,000

Region ID	Region	All trips	Completed	Cancelled	Fees (VND)
5	Nam Bộ	1	1	0	1,000,000

Region 5 has one completed trip here but none in C2. Its trip departed in December 2024, within the view's all-time scope and outside the quarterly count.

D3. Customer balances (v_debt)

Trip 1 shows the risk of joining a report directly to its receipts. Its confirmed fee is VND 1,000,000. Linking the report to both receipts would put that fee on two rows, so summing the joined rows would count it twice. v_debt first sums payments by trip_id. Each trip then contributes its fee once, alongside the total received for that trip, before the amounts are grouped by customer.

Customer ID	Customer	Confirmed fees (VND)	Receipts (VND)	Balance (VND)
1	Nguyễn Văn An	4,500,000	4,200,000	300,000
2	Trần Thị Bình	900,000	900,000	0
3	Lê Minh Châu	700,000	0	700,000
4	Phạm Quốc Dũng	1,100,000	1,100,000	0
5	Hoàng Thu Giang	2,300,000	2,300,000	0
6	Vũ Đức Hải	600,000	600,000	0
7	Đặng Mỹ Linh	1,000,000	1,000,000	0
8	Bùi Ngọc Mai	0	0	0
9	Đỗ Thanh Nam	0	0	0
10	Nguyễn Văn An	500,000	500,000	0

Reconciling fees and receipts

Confirmed fees of VND 11,600,000 less receipts of VND 10,600,000 leave VND 1,000,000 outstanding. Customer 1 owes the remaining VND 300,000 on trip 1. Customer 3 owes the full VND 700,000 on trip 6. These two balances account for the entire difference.

The 2025 receipts in C6 total VND 9,600,000. Adding the VND 1,000,000 received for trip 10 in December 2024 reconciles that figure to the all-time receipts in v_debt.